Sourav Ray

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Professional Summary

Senior Data & AI Engineer with an M.Tech in Operations Research and 3+ years of experience architecting high-performance distributed backend systems, data platform infrastructure, and scalable AI solutions. Proficient in Python, SQL, PySpark, AWS, and Snowflake, with proven expertise in code reviews, application architecture, total cost of ownership (TCO) optimization, and data quality governance. Technical Lead who has architected petabyte-scale ingestion engines and multi-agent platforms, achieving a 65% compute cost reduction and a 60% improvement in operational incident resolution. Recognized for evaluating emerging cloud technologies, mentoring engineering peers, and delivering resilient, low-latency enterprise solutions.

Technical Skills

Cloud & Data Platforms: AWS (S3, Lambda, Kinesis, Firehose, DMS, CloudWatch), Snowflake (Snowpark, Snowpipe, Cortex, Zero-Copy Cloning, Time Travel, Cost Optimization), GCP (Vertex AI, BigQuery).
Data Engineering: DBT, Apache Airflow, Apache Spark / PySpark, Coalesce, Apache Kafka, Apache Iceberg, SQL, ETL/ELT, Data Contracts, Data Mesh, Bronze/Silver/Gold Architecture.
AI / ML & GenAI: Large Language Models (LLMs), Agentic AI, Multi-Agent Orchestration, Retrieval-Augmented Generation (RAG), ChromaDB, Vector Search, Model Context Protocol (MCP), LangChain, LangGraph, TensorFlow, NLP, Prompt Engineering.
Programming & Frameworks: Python, SQL, Shell Scripting, FastAPI, Docker, Kubernetes, Git, GitHub Actions, CI/CD.
BI & Tools: Sigma, SAP BusinessObjects validation, Terraform, Linux.

Professional Experience

Factspan Analytics, Bengaluru, India
Senior Analyst / Data & AI Engineer Sep 2023 – Present

Customer 360 Data Mesh & Governance (Petabyte-Scale) | AWS, Snowflake, DBT, Python

- Architected a petabyte-scale Customer 360 data platform integrating 1,000+ sources across five ingestion patterns: RabbitMQ messaging queues, RDBMS databases, flat files, managed APIs, and real-time/event-based systems.
- Designed the end-to-end ingestion architecture: source systems → AWS ingestion services → Raw Kinesis → Lambda-based data-contract validation → validated Kinesis → Kinesis Firehose → S3 → Snowflake Snowpipe → Bronze/Silver/Gold transformation layers.
- Built a data-contract-driven validation framework that evaluated incoming records against schema and contract requirements before allowing data into the Bronze layer, detecting schema drift, missing mandatory columns, unexpected attributes, and invalid/null values for non-optional fields.
- Implemented fault-tolerant validation and recovery workflows where Lambda processing failures were routed to a messaging queue for controlled retry and reprocessing, preventing transient failures from causing data loss and enabling failed records to re-enter the validation stream.
- Led large-scale data modeling and transformation across approximately 100–200 DBT/SQL models spanning Bronze, Silver, and Gold layers for cross-domain Customer 360 data.
- Designed Bronze-layer transformations to flatten complex nested source payloads into downstream-consumable structures while preserving validated source data.
- Developed Silver-layer transformations for deduplication, enrichment, standardization, and business-level data preparation.

AI-Powered Network Incident Orchestrator | Python, MCP, Multi-Agent, Observability

- Owned the technical architecture for a multi-agent orchestration system coordinating specialized AI agents across ServiceNow, Infoblox, NetBox, SevOne, Datadog, Splunk, and LogicMonitor.
- Wrote and maintained the Python codebase for per-platform MCP servers; conducted design and code reviews with peers on style, testability, and efficiency.
- Designed a per-platform agent architecture, each backed by its own MCP server, providing a consistent abstraction layer over heterogeneous observability tools and device-management APIs, including secure SSH-based access to Cisco and non-Cisco network devices.
- Architected dynamic agent-selection and parallel-invocation logic correlating multi-source data (incidents, logs, metrics, topology, device health, configuration) into a single LLM-synthesized RCA report.

GenAI Schema Validator & Automated Column Mapping | Python, Vector Search, LLMs

- Architected an enterprise-scale schema and report-matching platform combining Exact Match, Semantic Match, and LLM-based inference layers — a RAG-style retrieval-and-reasoning pipeline applied to structured data for automated column mapping.
- Built responsible-AI governance controls to identify and exclude masked PII attributes from LLM inputs.
- Applied prompt engineering to auto-generate metadata for undocumented fields, reducing incorrect field mappings through structured validation.

Education

National Institute of Technology (NIT) Durgapur | M.Tech, Operations Research | CGPA: 8.36/10 2021 – 2023
Government College of Engineering and Textile Technology (GCETTB) | B.Tech 2015 – 2019

Certifications

- **Snowflake:** SnowPro Advanced Architect, SnowPro Advanced Data Scientist
- **AWS Cloud:** AWS Certified Machine Learning – Specialty, AWS Certified Solutions Architect – Associate